

Hypoxia and Altitude

Oxygen sensors (EPAS1, HIF1A), hemoglobin and erythropoiesis, obstructive sleep apnea, pulmonary edema of altitude (NOS3, EDN1), ECA and tolerance to altitude

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. There are no genetic signs of excess red blood cells: the predisposition to polycythemia vera is typical (percentile 24.2) and the oxygen sensors HIF1A (P582S C/C, A588T G/G) do not have the hyperactivity variants. The loci load of erythropoietin is at percentile 32.2.

To be considered. Genetic hemoglobin is low: the scores of hemoglobin, hematocrit and red blood cells are in the 11.7, 13.3 and 9.5 percentiles. This gives less oxygen transport reserve at altitude, but also less risk of thick blood in a long stay. In the vessels of the lung there are 3 slight alleles in favor of hypoxia vasoconstriction (ECA D 1 copy, NOS3 894T 1 copy, EDN1 198N 1 copy). Obstructive sleep apnea is typical in scores (75.2 and 79.6 percentiles), but the loci load is at percentile 85.0. The two markers of EPAS1 have the common genotype, without adaptation variants of high altitude people.

What does it mean? Genetic tolerance to medium altitude, without any strong marker. The practical points come from the drugs: dapagliflozin and acetazolamide together increase the risk of acidosis and dehydration at altitude, and tadalafil 5 mg daily already dilates a little the arteries of the lung.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 8 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 5 of the chip, 3 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 1,121,767 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 1941 loci, 1752 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the scores of sleep apnea and polycythemia vera, a higher percentile = more risk. In hemoglobin, hematocrit and red blood cells scores, the percentile only says if the level tends to be naturally higher or lower. There are no published polygenic scores for acute mountain malaise or pulmonary edema at altitude.

Sleep apnea

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Obstructive sleep apnea	PGS012528 (Huang 2026, LDpred, 307 thousand variants)	298,632 307,286 (weighted coverage 96%)	75.2	TYPICAL	typical
Obstructive sleep apnea	PGS003479 (Zhang 2022, LDpred2, 837 thousand variants)	821,693 836,839 (weighted coverage 97%)	79.6	TYPICAL	typical

Oxygen transport

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hemoglobin	PGS000168 (Vuckovic 2020, 515 variants)	424 515 (weighted coverage 83%)	11.7	LOW	without judgment
Hematocrit	PGS000167 (Vuckovic 2020, 502 variants)	414 502 (weighted coverage 84%)	13.3	LOW	without judgment
Red blood cells number	PGS000187 (Vuckovic 2020, 678 variants)	539 678 (weighted coverage 82%)	9.5	LOW	without judgment
Polycythemia vera	PGS001810 (Privé 2022, 67 variants)	65 67 (weighted coverage 99%)	24.2	TYPICAL	typical

4. Key variants by topic

Variants with effect described in the response to hypoxia. The adaptation variants to altitude are neither good nor bad out of altitude; there the judgment is "without judgment".

Oxygen sensors (via HIF)

When oxygen is lacking, HIF proteins are no longer degraded and bind hundreds of genes: EPO (more red blood cells), VEGF (more vessels), glycolysis and ventilation. EPAS1 (HIF-2 α) and EGLN1 (PHD2) are the genes of genetic adaptation of people living for millennia at high altitude, where variants stop the excess of red blood cells. HIF1A has two variants that make the protein more stable.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
EPAS1 (HIF-2 α)	rs4953354	intronic	A/A	imputation DR2=0.97	G (0 copy/s)	without judgment	EPAS1 (HIF-2 α): oxygen sensor that regulates EPO; G = adaptation allele of people living at high altitude (less hemoglobin in hypoxia); rare outside these populations	Moderate
EPAS1 (HIF-2 α)	rs13419896	intronic	G/G	imputation DR2=0.98	A (0 copy/s)	without judgment	Second sign of EPAS1: A = allele of adaptation to altitude (less hemoglobin in hypoxia)	Weak
HIF1A	rs11549465	P582S	C/C	chip	T (0 copy/s)	without judgment	P582S (T = Ser582): HIF-1 α more stable and more active in hypoxia; associated with better anaerobic performance and, in some studies, the higher risk of some cancers	Weak
HIF1A	rs11549467	A588T	G/G	chip	A (0 copy/s)	without judgment	A588T (A = Thr588): HIF-1 α more active	Weak

Hemoglobin, iron and erythropoiesis

Hemoglobin transports oxygen. Its level depends on EPO, iron (regulated by hepcidin, which TMPRSS6 controls) and hundreds of small-effect variants. At altitude, more hemoglobin is not always better: excess makes blood thick and associates with chronic mountain disease.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TMPRSS6	rs855791	V736A	A/G	chip	G (1 copy/s)	without judgment	TMPRSS6 V736A (regulates hepcidin): effect allele = more hemoglobin/iron [GWAS Catalog: Mean corpuscular hemoglobin, allele G, beta 0.142 [0.14-0.15] unit increase, p=8E-1100, PMID 34594039]	Strong

Pulmonary vessels and altitude

In hypoxia, the pulmonary arteries contract. When the contraction is strong and uneven, the pressure rises and the fluid comes out to the alveoli: it is the pulmonary edema of altitude. Nitric oxide (NOS3) dilates these arteries and endothelin-1 (EDN1) contract them. The ECA allele I is associated with better performance in altitude and resistance.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ACE (I/D, tag)	rs4343	I/D	G/A	chip	G (1 copy/s)	unfavourable (light)	ECA allele marker (G = D): more converting enzyme; allele I is associated with better performance in altitude and resistance	Weak

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NOS3	rs1799983	E298D	T/G	imputation DR2=1.00	T (1 copy/s)	unfavourable (light)	G894T (E298D; T = Asp298): less endothelial nitric oxide; associated with pulmonary edema in some studies	Weak
EDN1	rs5370	K198N	G/T	chip	T (1 copy/s)	unfavourable (light)	K198N (T = Asn198): plus endothelin-1 (pulmonary vasoconstriction); associated with pulmonary edema and pulmonary hypertension	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each trait, with each allele oriented towards the risk (in hemoglobin, in oxygen saturation and in erythropoietin, towards the highest level), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Sleep apnea	48 (33)	32 of 66	33	85.0	unfavourable
Oxygen saturation	2 (2)	2 of 4	2	90.3	without judgment
Hemoglobin	1875 (1703)	1671 of 3406	1699	47.1	without judgment
Erythropoietin	16 (14)	13 of 28	14	32.2	without judgment

6. Favourable

- **No predisposition to excess red blood cells:** polycythemia vera at percentile 24.2; loci load of erythropoietin in the 32.2. **Moderate.**
- **HIF1A without hyperactivity variants:** P582S C/C and A588T G/G (common genomes). **Weak.**
- **Obstructive sleep apnea in the scores:** percentiles 75.2 and 79.6. **Moderate.**

7. Unfavourable

- **Low genetic hemoglobin:** hemoglobin, hematocrit and red blood cells scores in the 11.7, 13.3 and 9.5 percentiles (the hemoglobin loci load is in the 47.1). Less oxygen reserve at altitude and maximum effort. Moderate (without judgment outside the altitude).
- **Lung vessels:** ECA I/D (marker G/A, I/D), NOS3 E298D T/G and EDN1 K198N G/T, each with 1 allele associated with more vasoconstriction in hypoxia and, in some studies, pulmonary edema of altitude. **Weak.**
- **Sleep apnea:** loci load at percentile 85.0. Weak (the most reliable scores are typical).
- **No judgment:** EPAS1 A/A and G/G (common genotypes without adaptation variants to altitude); TMPRSS6 V736A A/G (1 plus hemoglobin and iron allele).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Oxidative Phosphorylation Decline	Hypoxia reduces oxidative phosphorylation; no genetic markers of anomalous HIF response	Neutral
Mitochondrial dysfunction	HIF1A without hyperactivity variants; mitochondrial adaptation to hypoxia has not been read	Neutral
Exhaustion of Stem Cells	Erythropoiesis: low genetic hemoglobin (percentile 11.7) without polycythemia signal (percentile 24.2)	Neutral
Sleep disturbance	Typical obstructive apnea in scores (75.2 and 79.6 percentiles), loci load at percentile 85.0	Neutral
Vascular Aging	ECA D, NOS3 894T and EDN1 198N, 1 allele each: slightly favored pulmonary vasoconstriction	Unfavourable (mild)
Oxidative stress	Hypoxia and reoxygenation (including apnea) generate free radicals; without proper genetic marker	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Gradual rise:** above 2500 m, sleep at most 500 m above the previous night and make a day of rest every 3–4 days. It is the most effective measure against altitude evil and does not depend on genetics. *Suggestion (not applied).*
2. **Dapagliflozin at altitude:** dapagliflozin causes loss of water and sodium, and acetazolamide (the drug to prevent altitude sickness) causes mild metabolic acidosis. Together, they increase the risk of dehydration and ketoacidosis with normal glucose, especially with little food, long effort or ketosis. Before a trip above 2500–3000 m, talk to the doctor about suspending dapagliflozin 2–3 days before and during the stay. *Suggestion (not applied).*
3. **Acetazolamide:** If there is a rapid rise or history of bad altitude, the usual prevention dose is 125 mg 2x/day, starting the previous day. It is a non-antibiotic sulfonamide and needs a prescription. *Suggestion (not applied).*
4. **Tadalafil and pulmonary edema of altitude:** tadalafil is used to prevent pulmonary edema in those who had it, but at higher doses (10 mg 2x/day). The daily 5 mg gives some protection, without being studied for this purpose. Do not change the dose without the doctor; with the slight 3 alleles of the vessels of the lung, lack of air at rest or cough at altitude require to descend. *Suggestion (not applied).*
5. **Iron before a long stay at altitude:** The acclimatization needs iron. With low genetic hemoglobin, confirm that ferritin is not low before travel, keeping the target of ferritin below 100. Do not take iron without proven deficit. *Suggestion (not applied).*
6. **Sleep apnea:** The scores are typical; make a study of sleep (polygraph) only if there is strong snoring, pauses in the observed breathing or drowsiness. The altitude aggravates the periodic respiration of sleep; to sleep at altitude, melatonin 0.3–1 mg is preferable to sedatives. *Suggestion (not applied).*

Not recommended: benzodiazepines and other sedatives at altitude (depressing breathing); rising with symptoms of bad altitude; dexamethasone or nifedipine without medical indication; training in "intermittent hypoxia" without follow-up, with low genetic hemoglobin and dapagliflozin.

10. Limitations

- The adaptation variants to the altitude of EGLN1 (D4E and C127S) are not read with reliability in the data (one is not, the other is palindromic and imputed) and were removed.
- There are no polygenic scores for acute mountain malaise, pulmonary edema of altitude or cerebral edema of altitude. The candidate genes (ECA, NOS3, EDN1) have inconsistent results.
- The oxygen saturation loci load has only 2 loci and is not interpreted.
- Hemoglobin and red blood cell scores measure sea level; EPO response to hypoxia has little known genetic basis.
- The tolerance to altitude depends mainly on the speed of rise, previous acclimation, hydration, iron and drugs.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS012528 (Huang 2026) and PGS003479 (Zhang 2022) for obstructive sleep apnea, PGS000168, PGS000167 and PGS000187 (Vuckovic 2020) and PGS001810 (Privé 2022). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Beall 2010 and Yi 2010 (EPAS1), Lorenzo 2014 (EGLN1), Montgomery 1998 (ECA), studies of candidate genes of pulmonary edema at altitude (NOS3, EDN1) and the recommendations of Wilderness Medical Society 2024. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Oxygen sensors (via HIF)	EPAS1 (HIF-2 α)	rs4953354	A/A	imputation DR2=0.97	G	0	strand +	0 copy of the effect allele
Oxygen sensors (via HIF)	EPAS1 (HIF-2 α)	rs13419896	G/G	imputation DR2=0.98	A	0	strand +	0 copy of the effect allele
Oxygen sensors (via HIF)	HIF1A	rs11549465	C/C	chip	T	0	strand +	0 copy of the effect allele
Oxygen sensors (via HIF)	HIF1A	rs11549467	G/G	chip	A	0	strand +	0 copy of the effect allele
Hemoglobin, iron and erythropoiesis	TMPRSS6	rs855791	A/G	chip	G	1	strand +	1 copy of effect allele
Pulmonary vessels and altitude	ACE (I/D, tag)	rs4343	G/A	chip	G	1	strand +	1 copy of risk allele
Pulmonary vessels and altitude	NOS3	rs1799983	T/G	imputation DR2=1.00	T	1	strand +	1 copy of risk allele
Pulmonary vessels and altitude	EDN1	rs5370	G/T	chip	T	1	strand +	1 copy of risk allele